# AI-Based Image-to-Recipe Generation System Using CNN and Sequence Modeling

## Work Allocation Report

**Project-Based Placement Readiness Programme**
Training and Placement Cell

**Team ID:** Group_14
**Submission Phase:** Phase 2 – Requirement Analysis & Feasibility Review

## Abstract

This document presents the work allocation for the AI-Based Image-to-Recipe Generation System Using CNN and Sequence Modeling. Responsibilities have been distributed among the four team members according to their interests and technical expertise to ensure balanced participation throughout the project lifecycle. The allocation covers requirement analysis, literature survey, system design, model development, web application development, testing, documentation, and project coordination.

**Keywords:** Work Allocation, CNN, Recipe Generation, Deep Learning, Computer Vision, Project Management

## 1. Introduction

The Image-to-Recipe Generation System aims to automatically identify food items from uploaded images, predict ingredients using a Convolutional Neural Network (CNN), and generate cooking recipes using sequence modeling techniques. Responsibilities have been allocated to ensure efficient execution across research, documentation, development, testing, and coordination.

## 2. Team Composition

| Name | Primary Role |
|---|---|
| Abi Suneer | Research & Dataset Analyst |
| Akshai Murali | System Designer & Project Coordinator |
| Sethulakshmi C R | Requirement Analyst & Documentation Lead |
| Alen Joby | Web Development & Testing Engineer |

## 3. Detailed Work Allocation

### 3.1 Literature Survey and Dataset Research – Abi Suneer

Abi Suneer is responsible for conducting an extensive literature survey on image-based food recognition, ingredient prediction, and AI-based recipe generation techniques. Responsibilities include reviewing research papers, studying Food-101 and Recipe1M datasets, identifying limitations of existing approaches, documenting the research gap, and assisting in model evaluation.

### 3.2 Requirement Analysis and Documentation – Sethulakshmi C R

Responsible for preparing the Software Requirement Specification (SRS), identifying functional and non-functional requirements, documenting hardware and software requirements, preparing UML diagrams, and maintaining technical documentation.

### 3.3 System Design, AI Model Development and Project Coordination – Akshai Murali

Responsible for designing the overall architecture, selecting technologies, developing the CNN-based ingredient prediction module, integrating the recipe generation model, coordinating GitHub activities, monitoring milestones, and ensuring smooth integration.

### 3.4 Web Application Development and Testing – Alen Joby

Responsible for developing the Flask web application, implementing image upload, integrating frontend and backend, testing, debugging, evaluating performance, and improving user experience.

## 4. Work Allocation Summary

| Deliverable | Assigned To |
|---|---|
| Literature Survey | Abi Suneer |
| Requirement Analysis | Sethulakshmi C R |
| System Architecture & CNN | Akshai Murali |
| Frontend & Testing | Alen Joby |
| Final Review | All Team Members |

## 5. Coordination and Review Process

Weekly meetings will be conducted to review progress, resolve technical issues, and monitor milestones. All source code and documentation will be maintained through the team's GitHub repository. Each deliverable will be reviewed by all members before submission.

## 6. Schedule

Phase 2 activities including literature survey, requirement analysis, documentation, and conceptual design will be completed within the scheduled submission period through parallel collaboration.

## 7. Conclusion

The work allocation provides a balanced distribution of responsibilities and promotes effective collaboration, enabling successful completion of the AI-Based Image-to-Recipe Generation System.